Q1.a) Answer the following questions in two to three sentences:

i. What do you mean by denial of service attack? (1)

A Denial of Service (DoS) attack is an attempt to make a network service unavailable to its intended users by overwhelming the network with a flood of illegitimate requests, causing legitimate requests to be delayed or denied.

ii. What do you mean by rarest first in BitTorrent Networks? (2)

In BitTorrent networks, "rarest first" is a strategy used by peers to request the least available pieces of a file first. This ensures that rare pieces are propagated through the network early, improving the overall availability of the complete file among peers. **OR**

The rarest first strategy ensures that the pieces of a file that are least available in the network are prioritized for download. Peers will first request pieces that have the fewest copies available among the peers they are connected to.

iii. If a client wishes to access www.amazon.com. How this request will be processed by the hierarchy of DNS servers? (2)

When a client requests www.amazon.com, the query first goes to the local DNS resolver, which then queries the root DNS server. The root server directs the query to the .com top-level domain (TLD) server, which then directs it to Amazon's authoritative DNS server. The authoritative server provides the IP address for www.amazon.com, which is then returned to the client.

Q1.b)

i) Assuming no other traffic in the network, what is the throughput for the file transfer? (2.5)

The throughput is determined by the slowest link in the path. The slowest link is R1 with a rate of 500 kbps. Thus, the throughput for the file transfer is 500 kbps.

ii) Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B? (2.5

First, convert the file size to bits:

$$4 \text{ million bytes} = 4 \times 10^6 \times 8 \text{ bits} = 32 \times 10^6 \text{ bits}$$

The throughput is 500 kbps:

$$500 \text{ kbps} = 500 \times 10^3 \text{ bits per second}$$

Time to transfer the file:

$$\text{Time} = \frac{\text{File Size}}{\text{Throughput}} = \frac{32 \times 10^6 \text{ bits}}{500 \times 10^3 \text{ bits per second}} = 64 \text{ seconds}$$

Q1.c) When circuit switching is used, how many users can be supported? (5)

Each user requires 200 kbps when transmitting. The total link capacity is 10 Mbps:

$$10 \text{ Mbps} = 10 \times 10^3 \text{ kbps} = 10,000 \text{ kbps}$$

The number of users that can be supported:

$$\frac{10{,}000 \text{ kbps}}{200 \text{ kbps/user}} = 50 \text{ users}$$

Q2. a) Consider the provided figure and answer the following questions assuming TCP Reno is the protocol:

i. Slow Start Intervals: 0-6, 23-25.
ii. Congestion Avoidance Intervals: 6-16, 18-22.
iii. Loss Detection After 16th Round: Triple duplicate ACK.
iv. Loss Detection After 22nd Round: Timeout.
v. Initial ssthresh Value: 32

Q2.b) Compute the EstimatedRTT, DevRTT, and TCP TimeoutInterval after each SampleRTT value using given parameters:

Given:

- α = 0.125
- β = 0.25
- Initial EstimatedRTT = 100 ms
- Initial DevRTT = 5 ms
- SampleRTT values: 106 ms, 120 ms, 140 ms, 90 ms, 115 ms

Formulas:

$$\text{EstimatedRTT} = (1-\alpha) \times \text{EstimatedRTT} + \alpha \times \text{SampleRTT}$$

$$\text{DevRTT} = (1-\beta) \times \text{DevRTT} + \beta \times |\text{SampleRTT} - \text{EstimatedRTT}|$$

$$\text{TimeoutInterval} = \text{EstimatedRTT} + 4 \times \text{DevRTT}$$

2. For SampleRTT = 120 ms:

$$\text{EstimatedRTT} = (1-0.125) \times 100.75 + 0.125 \times 120 = 103.15625 \text{ ms}$$
$$\text{DevRTT} = (1-0.25) \times 5.3125 + 0.25 \times |120 - 103.15625| = 8.7109375 \text{ ms}$$
$$\text{TimeoutInterval} = 103.15625 + 4 \times 8.7109375 = 148 \text{ ms}$$

3. For SampleRTT = 140 ms:

$$\text{EstimatedRTT} = (1-0.125) \times 103.15625 + 0.125 \times 140 = 108.01171875 \text{ ms}$$
$$\text{DevRTT} = (1-0.25) \times 8.7109375 + 0.25 \times |140 - 108.01171875| = 14.61181640625 \text{ ms}$$
$$\text{TimeoutInterval} = 108.01171875 + 4 \times 14.61181640625 = 166.459 \text{ ms}$$

4. For SampleRTT = 90 ms:

$$\text{EstimatedRTT} = (1-0.125) \times 108.01171875 + 0.125 \times 90 = 105.26025390625 \text{ ms}$$
$$\text{DevRTT} = (1-0.25) \times 14.61181640625 + 0.25 \times |90 - 105.26025390625| = 13.77374267578125 \text{ ms}$$
$$\text{TimeoutInterval} = 105.26025390625 + 4 \times 13.77374267578125 = 160.355 \text{ ms}$$

5. For SampleRTT = 115 ms:

$$\text{EstimatedRTT} = (1-0.125) \times 105.26025390625 + 0.125 \times 115 = 106.84771728515625 \text{ ms}$$
$$\text{DevRTT} = (1-0.25) \times 13.77374267578125 + 0.25 \times |115 - 106.84771728515625| = 13.2919464111328125 \text{ ms}$$
$$\text{TimeoutInterval} = 106.84771728515625 + 4 \times 13.2919464111328125 = 159.015 \text{ ms}$$

Q3.a) Describe the concept of Software Defined Networks? How can we differentiate it with traditional network? (5)

Software Defined Networking (SDN) is an approach to networking that uses software-based controllers or application programming interfaces (APIs) to communicate with the underlying hardware infrastructure and direct traffic on the network. SDN decouples the network control plane from the data plane, allowing for more centralized and programmable network management.

Differentiation with Traditional Networks:

1. **Control and Data Planes:**
    - **SDN:** Control plane is separated from the data plane and is centralized.
    - **Traditional:** Control plane and data plane are integrated within the network devices (e.g., routers and switches).
2. **Configuration:**
    - **SDN:** Network configurations and management are more flexible and programmable.
    - **Traditional:** Configuration is typically static and manual.
3. **Scalability:**
    - **SDN:** Can dynamically adjust to changes in network demand.
    - **Traditional:** More challenging to scale and adapt to changes quickly.

Q3.b) Scheduling network packets using Priority service:

Given:

- Odd-numbered packets (High priority)
- Even-numbered packets (Low priority)

Transmission Order and Delays:

1. Packet 1 (High): Delay = 0 slots
2. Packet 2 (Low): Delay = 2 slot (waits for Packet 1)
3. Packet 3 (High): Delay = 0 slots
4. Packet 4 (Low): Delay = 5 slots (waits for Packet 1 and Packet 3)
5. Packet 5 (High): Delay = 0 slots
6. Packet 6 (Low): Delay = 5 slots (waits for Packet 1, Packet 3, and Packet 5)
7. Packet 7 (High): Delay = 1 slots
8. Packet 8 (Low): Delay = 4 slots (waits for Packet 1, Packet 3, Packet 5, and Packet 7)
9. Packet 9 (High): Delay = 0 slots
10. Packet 10 (Low): Delay = 3 slots (waits for Packet 1, Packet 3, Packet 5, Packet 7, and Packet 9)

11. Packet 11 (High): Delay = 0 slots
12. Packet 12 (Low): Delay = 3 slots (waits for Packet 1, Packet 3, Packet 5, Packet 7, Packet 9, and Packet 11)

Average Delay: 1.91

Q3.c) Datagram calculation for MP3 file:

Given:

- Datagram limit = 1,500 bytes (including header)
- IP header = 20 bytes
- MP3 file size = 74,000 bytes

Data payload per datagram:

$$\text{Data payload} = 1,500 - 20 = 1,480 \text{ bytes}$$

Number of datagrams:

$$\text{Number of datagrams} = \left\lceil \frac{74,000}{1,480} \right\rceil = \lceil 50 \rceil = 50$$

Q3.d) Longest-prefix matching:

- The datagram with destination address 11001000100100010101000101010101110010001 0010001 01010001 0101010111001000100100010101000101010101 will be forwarded to **Interface 3**.
- The datagram with destination address 11100001010000001100001100111100111000010 1000000 11000011 0011110011100001010000001100001100111100 will be forwarded to **Interface 2**

Q3.e) Calculate subnet bits, host per subnet, and subnet mask:

Networks	Subnet Bits	Host / Subnet	Subnet Mask
10.0.0.0/23	23	510	255.255.254.0
10.0.2.0/24	24	254	255.255.255.0
10.0.3.3/25	25	126	255.255.255.128
10.0.3.192/27	27	30	255.255.255.224
10.0.3.244/30	30	2	255.255.255.252

Q4.a) Using Dijkstra's algorithm, find the least cost path from source node u to all other destinations:

Q4:-

a)

Order	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)	D(z),p(z)
	(∞,-)	(∞,-)	(∞,-)	(∞,-)	(∞,-)
u	(7,u)	(9,u)	(6,u)		(∞,-)
ux	(7,u)	(9,u)		(9,x)	(∞,-)
uxv		(8,v)		(9,x)	(∞,-)
uxvw				(9,x)	(15,w)
uxvwy					(11,y)
uxvwyz					

Least-Cost Path Tree:

The resulting least-cost-path tree from uuu to all other nodes is as:

- $u \to v$ with cost 7
- $u \to x$ with cost 6
- $u \to x \to y$ with cost 9
- $u \to v \to w$ with cost 8
- $u \to x \to y \to z$ with cost 11

Q4.b) Border Gateway Protocol (BGP) questions:

i. Identify the routers, which are designated as eBGP routers?

eBGP routers are those that connect different autonomous systems (ASes). Typically, these are edge routers that handle inter-AS traffic.
1a, 1b, 1c, 2a, 3b,3d,4a,4b,4c,5a,5b,6a,6b,6e,7a,7b,7c

ii. Which autonomous systems are running iBGP?

iBGP runs within an autonomous system to exchange routing information between internal routers.
All AS except 200.

Q5.a) Services provided by the link layer:

1. **Framing:** The link layer encapsulates network layer datagrams into frames for transmission. It adds headers and trailers to the frames to enable error detection and handling.
2. **Error Detection and Correction:** The link layer provides mechanisms for detecting and correcting errors that may occur during data transmission. This ensures data integrity and reliability.

Q5.b) Channel partitioning methods:

1. Time Division Multiple Access (TDMA):

- **Pros:** Efficient use of the spectrum, avoids collisions.
- **Cons:** Requires synchronization, fixed time slots may be inefficient if users have variable data rates.

2. Frequency Division Multiple Access (FDMA):

- **Pros:** Simultaneous transmission possible, no synchronization required.
- **Cons:** Fixed frequency bands can be inefficient if demand is variable, limited number of users.

Q5.c) Cyclic Redundancy Check (CRC):

Given:

- Generator (G) = 1001
- Data payload (D) = 101110
- Number of bits in R (r) = 3

Solution:-

R=011